

Campus Complaint Portal: A Web-Based System for Efficient Complaint Management and Tracking

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Abstract—Educational institutions regularly receive complaints related to classrooms, laboratories, hostels, transportation, infrastructure, maintenance, and other campus facilities. Traditional complaint methods such as verbal communication, paper forms, and direct communication can make complaint tracking and record management difficult. The proposed Campus Complaint Portal provides a centralized web-based system for submitting, managing, assigning, and tracking complaints. Users can submit complaints by entering details such as category, title, description, location, and date. Administrators can review complaints, assign them to appropriate departments, and update their status. The system stores complaint information in a structured database, which supports organized information management [1]. The proposed system reduces manual work, improves complaint visibility, and provides a systematic method for managing campus complaints.

Index Terms—Campus Complaint Portal, Complaint Management, Complaint Tracking, Web Application, Student Services, Database Management

I. INTRODUCTION

Educational institutions regularly deal with different types of problems related to classrooms, laboratories, hostels, transportation, infrastructure, cleanliness, electricity, and maintenance. Students and staff need a simple way to report these problems and receive updates about their complaints.

In many institutions, complaints are still communicated through verbal communication, paper forms, telephone calls, emails, or direct communication with staff. These methods may make it difficult to maintain a centralized record and monitor the progress of complaints. A systematic software approach can help organize and manage such information effectively [2], [3].

The Campus Complaint Portal is proposed as a web-based platform that centralizes the complaint management process. Users can submit complaints through an online form, while administrators can review complaints, assign them to departments, and update their status. Database systems provide structured methods for storing and retrieving such information [1].

The system also allows users to track the current status of their complaints. This provides better visibility into the complaint process and reduces the need for repeated physical communication. The system follows structured software development and information management principles [2], [4].

II. PROBLEM STATEMENT

Traditional complaint management methods in educational institutions often depend on paper forms, verbal communication, emails, telephone calls, or direct communication. These methods can create difficulties in maintaining records, identifying the responsible department, monitoring pending complaints, and providing status information to users.

There is a need for a centralized system that allows students and staff to submit complaints online and enables administrators to manage complaints in an organized manner. A database-based system can provide structured storage and retrieval of complaint information [1].

The proposed Campus Complaint Portal addresses these problems by providing a single platform for complaint submission, department assignment, status updates, tracking, and record management.

III. OBJECTIVES

The main objectives of the Campus Complaint Portal are:

- To provide an online platform for submitting campus complaints.
- To categorize complaints based on their type.
- To maintain complaint records in a centralized database.
- To allow administrators to review and manage complaints.
- To assign complaints to the appropriate department.
- To allow users to track complaint status.
- To reduce manual complaint management.
- To improve transparency in the complaint handling process.
- To maintain pending and resolved complaint records.

These objectives are consistent with the use of information systems for supporting organizational activities and information management [4].

IV. LITERATURE SURVEY

Web-based information systems are commonly used to organize, process, and manage information. Systematic software development methods provide structured approaches for designing, implementing, and maintaining software systems [2], [3].

Object-oriented software development provides methods for identifying system objects, users, processes, and relationships.

These concepts can be useful in designing systems containing different users and functional modules [5].

Unified Modeling Language is also commonly used for representing software systems through different types of models. UML can help describe system structure and behavior in a clear way [6].

Database management is another important part of information-based systems. A structured database can be used to store user information, complaint details, department information, and status records [1].

Management information systems support organizational activities by collecting, processing, storing, and providing information to users and decision makers [4].

Based on these concepts, the proposed Campus Complaint Portal combines online complaint submission, centralized database storage, department assignment, complaint tracking, and administrative monitoring in one web-based system.

V. EXISTING SYSTEM

In the existing system, students and staff may report problems using paper forms, verbal communication, telephone calls, emails, or messages. The information may be maintained manually by different departments.

The major limitations of the existing system include:

- Manual maintenance of complaint records.
- Difficulty in tracking complaint status.
- No centralized complaint database.
- Difficulty in assigning complaints to departments.
- Delays in communication.
- Limited visibility for users.
- Difficulty in monitoring pending complaints.
- Increased administrative effort.

Such limitations can be reduced through a centralized information system that organizes records and provides access to relevant users [4].

VI. PROPOSED SYSTEM

The proposed Campus Complaint Portal is a centralized web-based system for managing campus complaints. Users can create an account and log in to the system before submitting a complaint.

A complaint contains information such as category, title, description, location, and date. The information is stored in the database for future retrieval and management [1].

After submission, the administrator reviews the complaint and assigns it to the appropriate department. The department can work on the complaint and update its status. The user can view the current status through the portal.

The complete complaint lifecycle includes submission, review, department assignment, processing, status updates, resolution, and closure. This structured approach supports systematic information management [2].

VII. SYSTEM REQUIREMENTS

A. Hardware Requirements

The basic hardware requirements are:

- Computer or laptop.
- Minimum 4 GB RAM.
- Minimum 10 GB available storage.
- Keyboard and mouse.
- Internet connection.

B. Software Requirements

The software requirements are:

- Windows or Linux operating system.
- HTML, CSS, and JavaScript.
- Node.js or equivalent backend technology.
- MySQL, MongoDB, or Supabase database.
- Visual Studio Code.
- Google Chrome or Microsoft Edge.

The use of appropriate software development tools and structured development methods supports organized implementation and maintenance [2], [3].

VIII. SYSTEM ARCHITECTURE

The Campus Complaint Portal follows a client-server architecture. The user interacts with the system through a web browser. The frontend sends requests to the backend, while the backend communicates with the database.

The administrator dashboard also communicates with the backend to retrieve complaint information, assign complaints, and update their status.

The separation of user interface, application processing, and data storage helps organize different responsibilities of the system [2], [3].

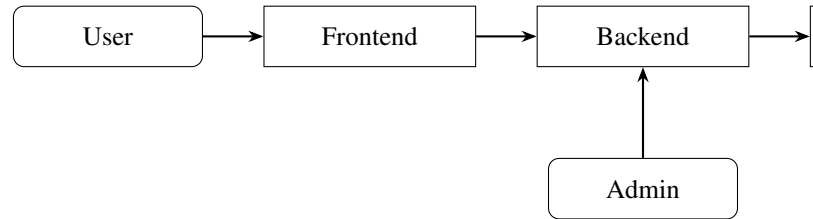


Fig. 1. System Architecture of Campus Complaint Portal

IX. SYSTEM WORKFLOW

The workflow of the proposed system consists of the following steps:

- 1) User opens the Campus Complaint Portal.
- 2) User registers or logs into the system.
- 3) User enters complaint details.
- 4) The system validates the complaint information.
- 5) Complaint information is stored in the database.
- 6) Administrator reviews the complaint.
- 7) Administrator assigns the complaint to the appropriate department.
- 8) Department works on the complaint.

- 9) Department updates the complaint status.
- 10) User tracks the complaint status.
- 11) The complaint is marked as resolved after completion.

This workflow provides a systematic sequence for managing information and activities within the system [5].

X. FLOWCHART

The main flow of the Campus Complaint Portal is shown below.

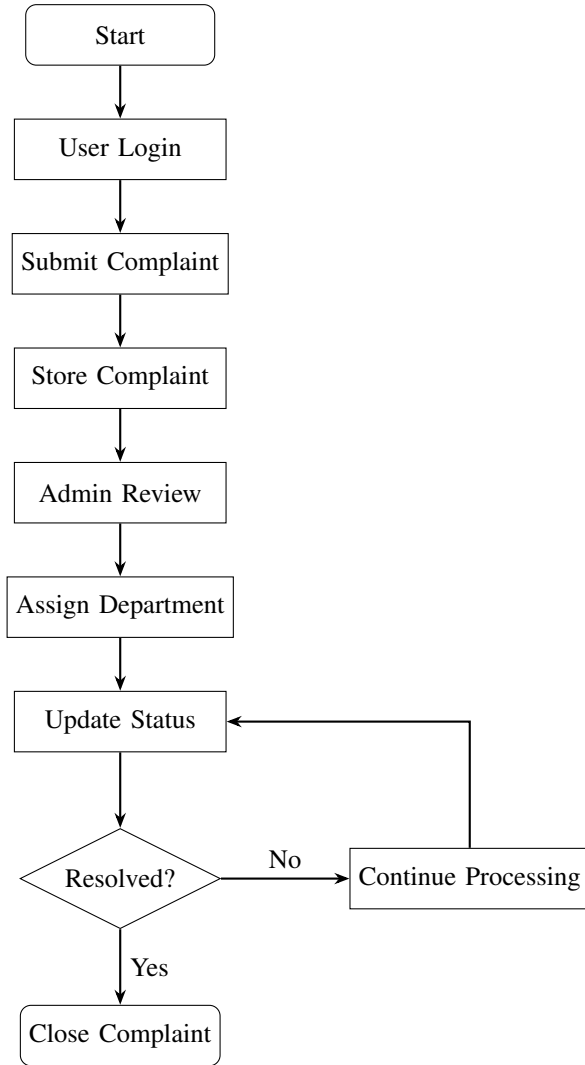


Fig. 2. Complaint Management Workflow

XI. MODULES

The proposed system consists of the following major modules.

A. User Module

The User Module allows students and staff to register, log in, submit complaints, and view their complaint status.

B. Complaint Submission Module

This module allows users to enter complaint information such as category, title, description, location, and date. The information is stored in the database.

C. Administrator Module

The Administrator Module allows administrators to view complaints, review complaint details, assign complaints to departments, and update complaint status.

D. Department Module

The Department Module allows the responsible department to view assigned complaints and update their progress.

E. Complaint Tracking Module

This module allows users to monitor the current status of their submitted complaints.

F. Notification Module

The Notification Module can be used to provide users with information about complaint status changes.

The modular organization of these functions follows the principle of separating system responsibilities into manageable components [2], [3].

XII. DATABASE DESIGN

The database is designed to store user and complaint information in an organized manner. Database systems provide mechanisms for storing, retrieving, and managing structured information [1].

A. User Table

TABLE I
USER TABLE

Field	Description
User ID	Unique user identifier
Name	Name of the user
Email	User email address
Password	Login password
Role	User or administrator

B. Complaint Table

TABLE II
COMPLAINT TABLE

Field	Description
Complaint ID	Unique complaint identifier
User ID	ID of the complaint creator
Category	Complaint category
Title	Complaint title
Description	Detailed complaint information
Location	Location of the problem
Date	Complaint submission date
Status	Current complaint status
Department	Assigned department

XIII. DATA FLOW DIAGRAM

The Data Flow Diagram represents the movement of information between users, the complaint portal, administrators, departments, and the database.

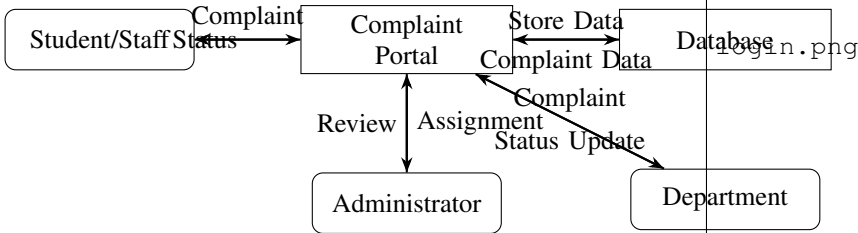


Fig. 3. Data Flow Diagram

XIV. IMPLEMENTATION

The Campus Complaint Portal is implemented using frontend, backend, database, and authentication components. The frontend provides the user interface for registration, login, complaint submission, and complaint tracking. The backend processes user requests and manages communication between the frontend and database.

The database stores user information, complaint details, department assignments, and complaint status. Authentication is used to control access to different parts of the system.

The implementation follows a structured software development approach in which different components perform specific responsibilities [2], [3].

XV. USER INTERFACE

The proposed system contains different interfaces for users and administrators.

A. Login Page

The login page allows registered users to enter their credentials and access the system.

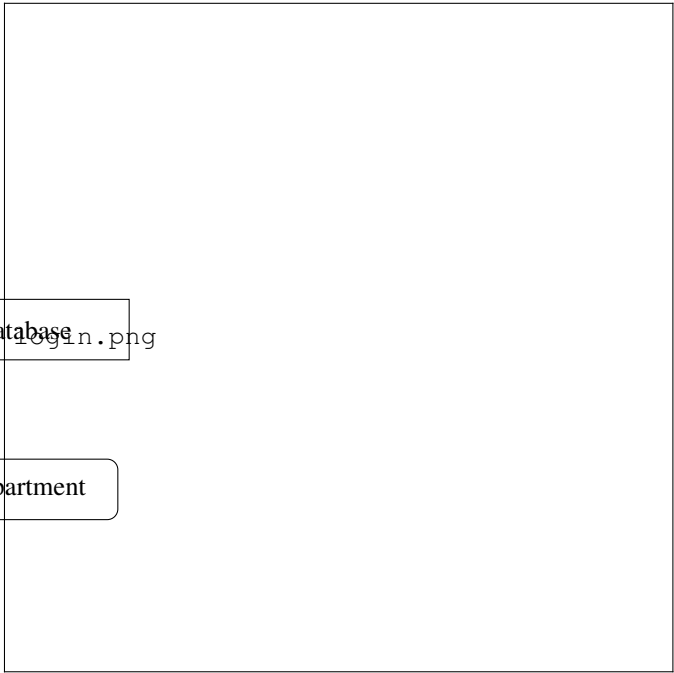


Fig. 4. Login Page

B. Complaint Form

The complaint form allows users to enter category, title, description, location, and other required information.

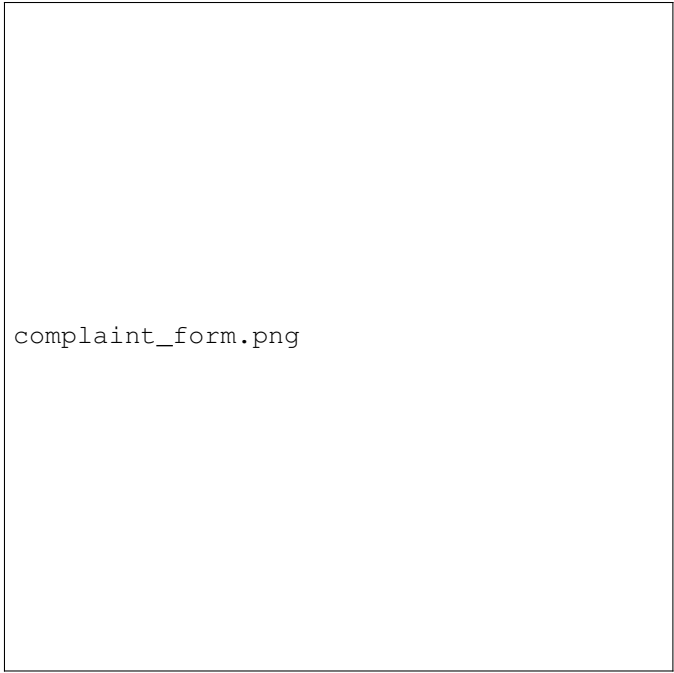


Fig. 5. Complaint Submission Form

C. User Dashboard

The user dashboard allows users to view their submitted complaints and check their current status.

user_dashboard.png

Fig. 6. User Dashboard

D. Admin Dashboard

The administrator dashboard provides functions for viewing, assigning, and updating complaints.

admin_dashboard.png

Fig. 7. Administrator Dashboard

XVI. TESTING

Testing is performed to verify that the main functions of the Campus Complaint Portal work correctly. Systematic testing

helps identify errors and verify whether the implemented functions satisfy the expected requirements [3].

TABLE III
SYSTEM TESTING

Test	Description	Result
Login	Verify user login	Pass
Registration	Verify new user registration	Pass
Complaint	Submit a complaint	Pass
Category	Select complaint category	Pass
Assignment	Assign complaint to department	Pass
Status	Update complaint status	Pass
Tracking	View complaint status	Pass
Resolution	Close resolved complaint	Pass

XVII. RESULTS AND DISCUSSION

The proposed Campus Complaint Portal provides a centralized platform for managing campus complaints. Users can submit complaints online without depending on paper-based methods.

The system stores complaint information in a database, allowing administrators to access and manage complaint records. Administrators can assign complaints to appropriate departments and update their status.

Users can track the progress of their complaints through the portal. This provides better visibility and reduces the need for repeated communication.

The centralized approach supports organized information management and can help administrators monitor pending and resolved complaints [1], [4].

XVIII. COMPARISON WITH TRADITIONAL SYSTEM

TABLE IV
TRADITIONAL SYSTEM VS PROPOSED SYSTEM

Feature	Traditional	Proposed
Complaint Submission	Manual	Online
Record Keeping	Manual	Database
Tracking	Limited	Available
Assignment	Manual	System Based
Status Visibility	Limited	Available
Centralized Records	No	Yes
Dashboard	No	Yes
Monitoring	Difficult	Easier

The comparison demonstrates how the proposed system organizes complaint information through a centralized information system [4].

XIX. ADVANTAGES

The major advantages of the proposed system are:

- Online complaint submission.
- Centralized complaint records.
- Easy complaint tracking.

- Reduced manual work.
- Organized department assignment.
- Better complaint status visibility.
- Easier administrative monitoring.
- Structured information management.

These features support the general objective of information systems to provide organized information for operational activities [4].

XX. LIMITATIONS

The current system has some limitations:

- Internet access is required.
- Complaint resolution depends on the responsible department.
- Users must provide accurate complaint information.
- Notification functionality may be limited.
- The system requires regular maintenance.

The effectiveness of an information system also depends on proper implementation, maintenance, and user interaction [2].

XXI. FUTURE SCOPE

The Campus Complaint Portal can be improved in the future by adding:

- Mobile application support.
- Email and SMS notifications.
- Automatic complaint prioritization.
- AI-based complaint categorization.
- Multilingual support.
- Complaint analytics and reports.
- Department performance analysis.
- User feedback and rating.
- Automatic complaint escalation.

These improvements can extend the information system and provide additional support for organizational decision-making and management [4].

XXII. CONCLUSION

The Campus Complaint Portal provides a centralized web-based solution for submitting, managing, assigning, and tracking campus complaints. The system reduces dependence on manual complaint management methods and provides better visibility of complaint status.

The use of a structured database allows complaint information to be stored and retrieved efficiently [1]. The administrator can review complaints, assign them to appropriate departments, and monitor their progress.

The proposed system follows systematic software development and information management principles [2], [3]. Future enhancements such as mobile applications, notifications, analytics, and intelligent complaint classification can further extend the functionality of the system.

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